Kennesaw State University 2026-2027 Undergraduate Catalog

Courses

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Data Science & Analytics

- ### DATA 1501: Introduction to Data Science

3 Credit Hours

Description
This course is intended to provide an introduction into the field of Data Science. Students will develop skills in appropriate technology and basic statistical methods by completing hands-on projects focused on real-world data and addressing the social consequences of data analysis and application.

- ### DATA 3010: Computer Applications of Statistics

3 Credit Hours

Prerequisite: STAT 1401 or ECON 2300 or STAT 3125 or STAT 2332 or MATH 3332
Description
This is an intermediate survey course of computer-based statistical software applications in the analysis and interpretation of data. Topics include developing a proficiency in coding in multiple languages through quantitative applications. Software packages include the most in-demand statistical languages and packages in the marketplace. (e.g. Python, SAS, R)

Course Learning Outcomes
Students who successfully complete this course will be able to:

1. Function as an informed user and critical consumer of statistics able to answer questions in an intelligent, informed, and insightful manner and pose intelligent questions concerning data, data analysis, and software use.
2. Obtain and correctly interpret the measures of central tendency and dispersion of a data set.

3. Generate and correctly interpret a variety of graphics for both univariate and multivariate analyses.
4. Generate random numbers and use them to perform Simple Random Sampling.
5. Generate and correctly interpret confidence intervals.
6. Represent themselves as proficient in each of these packages in a job interview or professional environment.
7. Identify types of statistical variables.
8. Differentiate the roles of explanatory and response variables.
9. State a proper research hypothesis.
10. Choose an appropriate statistical analysis for a given situation.
11. Summarize and communicate analysis results through a professional document.

- ## DATA 3230: Data Visualization

3 Credit Hours

Prerequisite: STAT 1401 or DATA 1501 or STAT 2332 or STAT 3125

Description

This course introduces students to the field of data visualization. The course covers basic design and evaluation principles to prepare and analyze large datasets, and standard visualization techniques for different types of data using modern data visualization software. The course prepares students to communicate clearly, efficiently, and in a visually compelling manner to a variety of audiences.

- ## DATA 3300: Data Science Ethics

3 Credit Hours

Prerequisite: STAT 3130

Description

As the field of data science and artificial intelligence continues to rapidly grow, so does the need for strong ethical guidelines. Throughout this course, students will learn the foundational ethical theories and frameworks, and the origins of ethics within data science. Students will use case studies to learn about the ethical dilemmas around the collection, management, and use of data, the use of models and algorithms, and the future of artificial intelligence and machine learning. Topics include Privacy, Informed Consent, Ownership, Security, Bias, Misinformation, Data Governance and Codes of Ethics.

- ## DATA 3396: Cooperative Study

1-3 Credit Hours

Prerequisite: Approval of the coordinator of cooperative education/internship.

Description

This course is a supervised work experience program for a minimum of two academic semesters at a site in business, industry, or government. It is for sophomore, junior, or senior-level students who wish to obtain successive on-the-job experience in conjunction with their academic training.

- ## DATA 3398: Internship

1-9 Credit Hours

Prerequisite: Approval of the program coordinator and department chair.
Description
This course is a supervised, credit-earning work experience of one academic semester with a previously approved business firm, private agency, or government agency.

- ## DATA 4000: Data Science Communication

3 Credit Hours

Prerequisite: STAT 2332 or (DATA 3010 and (STAT 3120 or STAT 3125))
Description
This course equips students to orally communicate data analysis results adapted to both technical and non-technical audiences. Students learn and practice essential data presentation skills, such as using narratives and visuals to communicate data analysis insights for solving business problems.

- ## DATA 4030: Programming in R

3 Credit Hours

Prerequisite: DATA 3010 or STAT 3125
Description
In this course, students will learn R programming to effectively manage, explore, visualize, and present data. The course covers practical statistical-computing issues, including reading, manipulating, and analyzing data, using control statements, existing functions, and user-created functions. Reporting results using R packages, such as R Markdown, may also be covered.

- ## DATA 4130: Nonparametric Methods

3 Credit Hours

Prerequisite: STAT 3130

Description

This course focuses on distribution-free statistical inference techniques applicable in scenarios where parametric assumptions may not hold. Students will extend their foundational knowledge by exploring nonparametric methodologies for hypothesis testing and estimation in a variety of data scenarios. More advanced computational and statistical techniques may also be covered at the instructor's discretion.

Course Learning Outcomes

Students who successfully complete this course will be able to:

1. Clearly articulate the advantages and disadvantages of nonparametric methods as compared to parametric methods.
2. Correctly implement and interpret one-population nonparametric methods.
3. Correctly implement and interpret two-population nonparametric methods.
4. Correctly implement and interpret nonparametric methods for more than two populations.
5. Correctly implement and interpret nonparametric methods for evaluating the association between variables.

- ## DATA 4140: Python for Data Science

3 Credit Hours

Prerequisite: DATA 3010 and (STAT 3130 or IET 3403 or ISYE 3600)

Description

This course introduces students to analytic methods using Python. The core focus is the development of Python knowledge within an analytic model development focus. Students will learn fundamental data structures, key algorithms and their application in applying analytic/machine learning methodologies.

- ## DATA 4310: Statistical Data Mining

3 Credit Hours

Prerequisite: STAT 3130 or permission of the instructor.

Description

Data Mining is an information extraction activity whose goal is to discover hidden facts contained in databases, perform prediction and forecasting, and generally improve their performance through interaction with data. The process includes data selection, cleaning, coding, using different statistical, pattern recognition and machine learning techniques, and reporting and visualization of the generated structures. The course will cover all these issues and will illustrate the whole process by examples of practical applications. The students will use recent SAS Enterprise Miner software.

- ## DATA 4330: Applied Binary Classification

3 Credit Hours

Prerequisite: STAT 4210

Description

Common applications of binary classification include credit worthiness and the associated development of a credit risk score, fraud detection, and the presence of a disease. Students will learn to use logistic regression, odds, ROC curves, and maximization functions to apply binary classification concepts to real-world datasets. This course utilizes statistical coding software and students are expected to have an advanced knowledge of this software.

- ## DATA 4400: Directed Study

3 Credit Hours

Prerequisite: Approval of the instructor, major area committee, and department chair.

Description

Special advanced topics external to regular course offerings.

- ## DATA 4490: Special Topics in Statistics

3 Credit Hours

Prerequisite: STAT 3130

Description

Special topics of interest to faculty and students.

- ## DATA 4990: Data Science Capstone

3 Credit Hours

Prerequisite: STAT 4210

Description

Capstone projects challenge students to acquire and analyze data to solve real-world problems. Students will have to synthesize and strengthen the knowledge and skills learned through the program such as data visualization, inference and modeling, data wrangling, data organization, data mining and machine learning, as well as storytelling with the data.

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